MATH 243 — Probability and Statistics

Probability Basics: Sets, Counting, Rules, Odds and Joint Tables

Lesson notes · Exam approach · TF practice with solutions · Quick notes (KKS Sir's portion — Probability & ANOVA)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

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Lesson 2 — Counting: Permutations, Combinations, Classical Probability — nPr, nCr, committees, cards, balls
Lesson 3 — Multiplication Rule, Independence and Odds — independent vs dependent, circuits, odds
Lesson 4 — Joint Probability and the 2 × 2 Table — joint/marginal table, newspaper and machine problems
Quick Notes — Probability Basics — revision bullets

> **NOTE** Page numbers refer to the KKS Sir lecture PDFs: Lecture 1.pdf (6 pp.), Lecture 2.pdf (9 pp.) and Lecture 3 & 4.pdf (12 pp.), written as Lec 1 p. 3 etc. "TF p." is the page of MATH 243 All Merged Questions.pdf; "22 batch p." is the page of MATH 243 (22 batch).pdf (the 2023-24 paper).

Exam Map — What the TF Papers Ask

Session (paper)	Questions answered by this note [lesson]
2023-24 (MATH 243) TF p. 22 batch p.1	Q1(a) Can two events be mutually exclusive and independent at the same time? Justify (6) [L3]
2016-17 (MATH 245) TF p. 20	Q7(a) Circuit of 4 parallel paths (0.02; 0.01–0.01; 0.01–0.01; 0.02 failure probabilities): P(circuit does not operate) (10) [L3]
2014-15 (MATH 241) TF p. 11	Q8(a) Box of 8 red, 3 white, 9 blue; 3 drawn: P(all red), P(2 red 1 white), P(at least 1 white) (12) [L2]
2013-14 (MATH 241) TF p. 8	Q7(b)(i) 3 cards drawn without replacement: P(all from different suits) (7) [L2]
2011-12 (MATH 241) TF p. 2	Q7(b)(ii) A, B, C solve a problem with probabilities 3/7, 2/3, 4/5 independently — P(problem solved) (8) [L3]

EXAM FOCUS 5 questions (43 marks) in 11 papers. These are short parts (6–12 marks) that usually open Section A: counting with nCr, P(at least one) = 1 − P(none), and the question "can two events be both mutually exclusive and independent?" (2023-24). Learn the complement trick and the series/parallel rule first; they also feed every later lesson.

Lesson 1 — Sample Space, Events and the Probability Rules

Lec 1 pp. 2–6 · Lec 2 pp. 1–2

1. What Probability Measures (Lec 1 p. 2)

Probability measures uncertainty. It is a number between 0 and 1: 0 means the event cannot happen, 1 means it certainly happens. We use it when the exact outcome of an experiment cannot be predicted but the set of possible outcomes is known.

Experiment	Event	Probability
Toss a fair coin	Head	1/2 = 0.5
Roll a die	a 4 / an even number	1/6 / 3/6 = 0.5
Class of 30 (12 girls, 18 boys)	select a girl / a boy	12/30 = 0.4 / 18/30 = 0.6
MCQ with 4 options, guessing	correct / wrong	1/4 = 0.25 / 3/4 = 0.75

2. Random Experiment, Sample Space and Events (Lec 2 p. 1, Lec 1 pp. 2–3)

A **random experiment** has three properties: (a) all possible outcomes are known in advance; (b) the outcome of any one performance is not known in advance; (c) it can be repeated under identical conditions (coin toss, die roll, drawing a card, choosing a student at random).

- **Sample space S** — the set of all possible outcomes. Die: S = {1, 2, 3, 4, 5, 6}; card: the 52 cards; baby's sex: {Male, Female}; coin: {H, T}.
- **Event** — any subset of S. Even number on a die: A = {2, 4, 6}; head: B = {H}; a king: C = {K♥, K♦, K♣, K♠}.
- **Equally likely events** — events with the same chance of occurring (head and tail of a fair coin; faces 1 and 2 of a fair die).

3. Set Operations (Lec 1 pp. 3–5)

With S = {1, …, 6}, A = {2, 4, 6} (even) and B = {1, 2} (less than 3):

Operation	Meaning	Example
$A \cup B$	A or B or both	{1, 2, 4, 6}
$A \cap B$	A and B together	{2}
A^c	everything in S except A	{1, 3, 5}
∅	empty set (impossible event)	rolling a 7
Disjoint (mutually exclusive)	$A \cap B = \emptyset$ — cannot happen together	{1} and {3}; {1, 2} and {3, 4}

De Morgan's laws (Lec 2 p. 2)

$$(A \cup B)^c = A^c \cap B^c$$
$$(A \cap B)^c = A^c \cup B^c$$

Example 1: S = {1, …, 6}, A = {1, 2, 3, 4}, B = {3, 4, 5, 6}. $A \cup B = S$, so $(A \cup B)^c = \emptyset$; check: $A^c = \{5, 6\}$, $B^c = \{1, 2\}$, $A^c \cap B^c = \emptyset$ ✓. Example 2: S = {a, b, c, d, e}, A = {a, b, c}, B = {b, d}: $A \cap B = \{b\}$, $(A \cap B)^c = \{a, c, d, e\}$; $A^c \cup B^c = \{d, e\} \cup \{a, c, e\} = \{a, c, d, e\}$ ✓.

4. The Axioms and the Addition Rule (Lec 1 pp. 4–6)

1. **Non-negativity:** P(A) ≥ 0 (P(7 on a die) = 0, never negative).
2. **Total probability:** P(S) = 1 (P(1) + … + P(6) = 1).
3. **Additivity for disjoint events:** P(A ∪ B) = P(A) + P(B) when A ∩ B = ∅ (P(1 or 2) = 1/6 + 1/6 = 2/6).

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$P(A^c) = 1 - P(A)$$

Die: P(A ∪ B) = 3/6 + 2/6 − 1/6 = 4/6. Cricket (Lec 1 p. 6): P(play) = 0.4, P(watch) = 0.7, P(both) = 0.35 → P(play or watch) = 0.4 + 0.7 − 0.35 = 0.75.

TRAP Add probabilities only for disjoint events. For overlapping events the intersection is counted twice, so subtract it once. "At least one" is almost always easier as 1 − P(none).

Exam Approach

1. Write S and the event as sets (or as counts n(A), n(S)).
2. Decide: union (or) → addition rule; intersection (and) → multiplication rule (Lesson 3); "not" or "at least one" → complement.
3. Use De Morgan to turn P(neither) into 1 − P(A ∪ B).

Practice — Lesson 1

(a) Adapted — Lec 1 p. 6. (b) Adapted — Lec 3 & 4 p. 7. (c) Constructed.

(a) 40 % of students play cricket, 70 % watch it and 35 % do both. What percentage do neither?

(b) 60 % study Mathematics, 50 % Physics, 30 % both. Find P(M or P), P(M but not P), P(neither).

(c) Show with De Morgan that $P(A^c \cap B^c) = 1 - P(A) - P(B) + P(A \cap B)$.

✓ Solution

(a) P(play ∪ watch) = 0.4 + 0.7 − 0.35 = 0.75, so P(neither) = 1 − 0.75 = 0.25 → 25 %.

(b) P(M ∪ P) = 0.6 + 0.5 − 0.3 = 0.8; $P(M \cap P^c) = P(M) - P(M \cap P) = 0.6 - 0.3 = 0.3$; P(neither) = 1 − 0.8 = 0.2.

(c) $A^c \cap B^c = (A \cup B)^c$, so $P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - [P(A) + P(B) - P(A \cap B)]$.

Lesson 2 — Counting: Permutations, Combinations, Classical Probability

Lec 2 pp. 2–8

1. Permutations — Order Matters (Lec 2 pp. 2–3)

$$nPr = n! / (n - r)!$$

Example	Working	Answer
Arrange 3 books A, B, C	3P3 = 3!/0!	6 (ABC, ACB, BAC, BCA, CAB, CBA)
Select and arrange 2 of 4 students	4P2 = 4!/2!	12
President, VP, secretary from 5 members	5P3 = 5!/2!	60

2. Combinations — Order Does Not Matter (Lec 2 pp. 3–5)

$$nCr = n! / (r! (n - r)!)$$

Problem	Working	Answer
Choose 2 of 3 books	3C2	3
Committee of 3 from 5	5C3	10
3 men from 8 and 2 women from 12	8C3 × 12C2 = 56 × 66	3696
5 from 6 men + 4 women, at least 1 woman	10C5 − 6C5 = 252 − 6	246
5 from 6 M + 4 W, ≥ 1 woman and ≥ 2 men	6C2·4C3 + 6C3·4C2 + 6C4·4C1 = 60 + 120 + 60	240
5-card hands with exactly one ace	4C1 × 48C4 = 4 × 194580	778320

MEMORIZE "Independent choices multiply" (men AND women: 8C3 × 12C2). "At least" = total − the complement case, or add the separate splits.

3. Approaches to Probability (Lec 2 p. 5)

- **Classical** — equally likely outcomes: P(A) = n(A)/n(S).
- **Relative frequency** — long-run proportion of times A occurs in repeated trials.
- **Subjective** — a personal degree of belief (used when no repetition is possible).

$$P(A) = n(A) / n(S) \quad \text{(classical)}$$

4. Worked Examples (Lec 2 pp. 5–8)

- Heart from 52 cards: 13/52 = 1/4.
- Bag 4 white, 6 red: P(R) = 6/10 = 3/5, P(W) = 4/10 = 2/5 → not equally likely.
- Die: P(even) = 3/6, P(> 4) = 2/6, P(odd and < 4) = 2/6; even and odd are equally likely (both 1/2).
- Two children S = {BB, BG, GB, GG}: P(both boys) = 1/4, P(at least one girl) = 3/4, P(at most one boy) = 3/4, P(exactly one girl) = P(exactly one boy) = 1/2, P(same sex) = 1/2.

Exam Approach

1. Decide whether order matters (arrangement → P; selection → C).
2. Count the sample space: n(S) = NCr for drawing r items at once.
3. Count the favourable cases as a product of C's, one per colour/type.
4. For "at least one", use 1 − P(none). Simplify the fraction and give a decimal.

Practice — Lesson 2

(a) Actual — 2014-15 Q8(a) (TF p. 11). (b) Actual — 2013-14 Q7(b)(i) (TF p. 8). (c) Adapted — Lec 3 & 4 p. 7.

(a) A box contains 8 red, 3 white and 9 blue balls. Three balls are drawn at random. Find the probability that (i) all 3 are red, (ii) 2 are red and 1 is white, (iii) at least 1 is white. (12)

(b) 3 cards are drawn one after another without replacement from a deck of 52. Find the probability that they come from different suits. (7)

(c) A bag has 3 red, 4 green and 5 blue balls; two are drawn. Find P(both red), P(one red one green), P(at least one blue).

✓ Solution

(a) Total balls 20, n(S) = 20C3 = 1140.

```
(i)   P(3 red)         = 8C3 / 20C3           = 56/1140  = 14/285  = 0.0491
(ii)  P(2 red, 1 white) = 8C2 × 3C1 / 20C3   = 84/1140 = 7/95   = 0.0737
(iii) P(no white)      = 17C3 / 20C3          = 680/1140 = 34/57
      P(at least 1 white) = 1 − 34/57 = 23/57 = 0.4035
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(b) Sequential multiplication rule: the first card can be anything (52/52); the second must avoid the first card's suit (39 of the remaining 51); the third must avoid both suits (26 of 50).

$$\mathbf{P = (52/52)(39/51)(26/50) = 169/425 = 0.3976}$$

(c) 12C2 = 66. P(both red) = 3C2/66 = 3/66 = 1/22; P(one red, one green) = (3C1 × 4C1)/66 = 12/66 = 2/11; P(at least one blue) = 1 − 7C2/66 = 1 − 21/66 = 15/22.

Lesson 3 — Multiplication Rule, Independence and Odds

Lec 2 pp. 8–9 · Lec 3 & 4 pp. 1–6

1. Multiplication Rule (Lec 2 pp. 8–9)

Independent: $P(A \cap B) = P(A) \cdot P(B)$
Dependent: $P(A \cap B) = P(A) \cdot P(B \mid A)$

Coin and die: $P(H \cap 4) = (1/2)(1/6) = 1/12$ — the same as counting 1 of the 12 outcomes. Without replacement (3 red, 2 blue): P(red then red) = (3/5)(2/4) = 3/10 = 3C2/5C2. Two aces without replacement: (4/52)(3/51) = 1/221.

2. Independent vs Mutually Exclusive ★ (textbook, asked 2023-24 Q1(a))

	Mutually exclusive (disjoint)	Independent
Definition	$A \cap B = \emptyset$, so $P(A \cap B) = 0$	$P(A \cap B) = P(A)P(B)$; $P(B \mid A) = P(B)$
Meaning	if A happens, B cannot	A happening does not change the chance of B
Addition	$P(A \cup B) = P(A) + P(B)$	$P(A \cup B) = P(A) + P(B) - P(A)P(B)$

EXAM-READY ANSWER Two events with positive probabilities cannot be both mutually exclusive and independent. If they are mutually exclusive, $P(A \cap B) = 0$; if independent, $P(A \cap B) = P(A)P(B) > 0$ when P(A), P(B) > 0. The two conditions agree only if P(A) = 0 or P(B) = 0. Intuitively, disjoint events are strongly dependent: knowing A happened tells us B did not, so $P(B \mid A) = 0 \neq P(B)$. Example: one die, A = {1}, B = {2}: $P(A \cap B) = 0$ but $P(A)P(B) = 1/36$.

3. Several Independent Events — P(at least one) and Circuits (textbook)

$P(\text{at least one of } A_1, \ldots, A_k) = 1 - \Pi\,(1 - P(A_i))$
Series: works if all work · Parallel: fails only if all branches fail

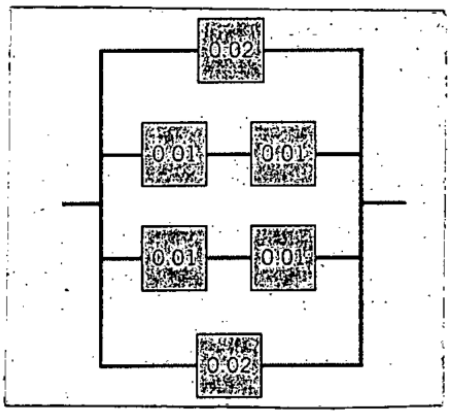


Figure 3.1 — Circuit of 2016-17 Q7(a): four parallel paths with failure probabilities (TF p. 20)

4. Odds (Lec 3 & 4 pp. 1–2)

Odds in favour of $A = P(A) / P(A^c) = n(A) : n(A^c)$
If odds in favour are a : b, $P(A) = a/(a + b)$

- Odds in favour and odds against are reciprocals.
- Red:blue odds 3 : 7 → P(red) = 3/10 = 0.3, P(blue) = 0.7. Odds 5 : 3 → P(A) = 5/8.
- P(win) = 0.2 → odds against = 0.8/0.2 = 4 : 1 (for every 1 chance of winning there are 4 of losing).

Worked example — X and Y solve a problem (Lec 3 & 4 pp. 2–5)

Odds against X are 8 : 6 → P(X) = 6/14 = 3/7, $P(X^c)$ = 4/7. Odds in favour of Y are 14 : 16 → P(Y) = 7/15, $P(Y^c)$ = 8/15.

Asked	Independent case	Dependent case, P(Y | X) = 0.6
Both solve	(3/7)(7/15) = 1/5	(3/7)(3/5) = 9/35
At least one (= either)	1 − (4/7)(8/15) = 73/105	3/7 + 7/15 − 9/35 = 67/105
Neither	(4/7)(8/15) = 32/105	1 − 67/105 = 38/105
Exactly one	73/105 − 21/105 = 52/105	67/105 − 27/105 = 40/105

TRAP In the dependent case you may not multiply $P(X^c)P(Y^c)$ — use the complement of the union. "Odds against 8 : 6" means P(fail) = 8/14, not 8/6.

Exam Approach

1. Turn odds into probabilities first: a : b → a/(a + b).
2. Independent → multiply; at least one → 1 − product of failures.
3. Circuits: series branch works with Π p_{works}; a parallel block fails with Π q_{fails}.
4. Dependent → use P(A)P(B | A) and the addition rule.

Practice — Lesson 3

(a) Actual — 2023-24 Q1(a) (22 batch p. 1). (b) Actual — 2011-12 Q7(b)(ii) (TF p. 2). (c) Actual — 2016-17 Q7(a) (TF p. 20).

(a) Can two events be mutually exclusive and independent simultaneously? Justify your answer. (6)

(b) The probabilities that A, B and C can solve a problem are 3/7, 2/3 and 4/5. If all try independently, find the probability that the problem will be solved. (8)

(c) The circuit (Figure 3.1) operates iff there is a path of functional devices from left to right. Devices fail independently with the probabilities shown. What is the probability that the circuit does not operate? (10)

✓ Solution

(a) No (unless one event has probability 0) — model answer in the callout of §2.

(b) Solved means at least one solves it.

$$\mathbf{P = 1 - (1 - 3/7)(1 - 2/3)(1 - 4/5) = 1 - (4/7)(1/3)(1/5) = 1 - 4/105 = 101/105 \approx 0.9619}$$

(c) The four paths are in parallel, so the circuit fails only if every path fails.

```
Path 1 (one device):  P(fail) = 0.02
Path 2 (two in series): works only if both work = 0.99 × 0.99 = 0.9801 → P(fail) = 1 − 0.9801 = 0.0199
Path 3 (two in series): P(fail) = 0.0199
Path 4 (one device):  P(fail) = 0.02
P(circuit does not operate) = 0.02 × 0.0199 × 0.0199 × 0.02 = 1.5840e-07
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$$\mathbf{P(\text{not operate}) \approx 1.584 \times 10^{-7}}$$

Lesson 4 — Joint Probability and the 2 × 2 Table

Lec 3 & 4 pp. 6–12

1. Joint Probability (Lec 3 & 4 p. 6)

Joint probability P(A ∩ B) is the probability that A and B happen at the same time: P(A)P(B) if independent, P(A)P(B | A) (or P(B)P(A | B)) if dependent.

2. The Joint Probability Table (Lec 3 & 4 p. 8)

```
            Y = yes        Y = no         Total
X = yes   P(X ∩ Y)     P(X ∩ Y^c)      P(X)
X = no    P(X^c ∩ Y)   P(X^c ∩ Y^c)    P(X^c)
Total       P(Y)         P(Y^c)          1
```

Each cell is a joint probability; the row and column totals are the marginal probabilities. Fill the table from the given numbers, then read any answer from it. The same table can hold counts n(·) that are divided by n(S).

3. Worked Problems (Lec 3 & 4 pp. 8–11)

Problem 1 — newspapers: 100 employees, 75 read English, 50 Bangla, 40 both

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            E      E^c    Total
B          40      10      50
B^c        35      15      50
Total      75      25     100
```

(a) P(E) = 0.75; (b) P(E ∪ B) = 0.75 + 0.50 − 0.40 = 0.85; (c) none: 1 − 0.85 = 0.15; (d) Bangla only: (50 − 40)/100 = 0.10.

Problem 2 — ring and necklace

P(R) = 0.20, P(N) = 0.30, P(neither) = 0.60: P(R ∪ N) = 1 − 0.60 = 0.40; P(R ∩ N) = 0.20 + 0.30 − 0.40 = 0.10.

Problem 3 — 10 men and 20 women, half of each with brown eyes

n(M ∩ B) = 5, n(B) = 15: P(M ∪ B) = 10/30 + 15/30 − 5/30 = 20/30 = 2/3. (The slide writes P(M) as 1/10 by a slip; it is 10/30 = 1/3, and the final 2/3 is correct.)

Problem 4 — machines A (40 %) and B (60 %), defective 5 % and 10 %

```
              Defective (Y)     Good (Y^c)       Total
Machine A   0.4×0.05 = 0.02   0.4×0.95 = 0.38   0.40
Machine B   0.6×0.10 = 0.06   0.6×0.90 = 0.54   0.60
Total          0.08              0.92           1
```

P(defective) = 0.08; P(A | defective) = 0.02/0.08 = 0.25; P(good and from B) = 0.54. This is already Bayes' theorem (Lesson 6) in table form.

> **NOTE** Problem 5 of the lecture (two pens from 3 blue, 2 red, 3 green → joint pmf of X = blue, Y = red) is the discrete joint distribution of Lesson 10 in the Joint Distributions note; 2023-24 Q2(b) asks the same problem with red/blue/black pens.

Exam Approach

1. Draw the 2 × 2 table with totals; put the given numbers in.
2. Fill the gaps by subtraction (each row and column must add up).
3. Answer "or" with the addition rule, "only" with a single cell, "given" with cell ÷ total.

Practice — Lesson 4

(a) Constructed. (b) Adapted — Lec 3 & 4 p. 10.

(a) In a class 45 % like tea, 35 % like coffee and 25 % like neither. Find P(both) and P(tea only).

(b) Machine A makes 30 % and B 70 % of items; 4 % of A's and 2 % of B's items are defective. Build the joint table and find P(defective) and P(A | defective).

✓ Solution

(a) $P(T \cup C) = 1 - 0.25 = 0.75$; $P(T \cap C) = 0.45 + 0.35 - 0.75 = 0.05$; P(tea only) = $0.45 - 0.05 = 0.40$.

(b) Cells: $A \cap D = 0.3 \times 0.04 = 0.012$, $A \cap D^c = 0.288$, $B \cap D = 0.7 \times 0.02 = 0.014$, $B \cap D^c = 0.686$. $P(D) = 0.012 + 0.014 = 0.026$; $P(A \mid D) = 0.012/0.026 = 0.4615$.

Quick Notes — Probability Basics

Revision points for Lessons 1–4.

L1 · Events and Rules

- Random experiment: outcomes known, result unknown, repeatable. S = all outcomes; event $\subseteq$ S.
- De Morgan: $(A \cup B)^c = A^c \cap B^c$; $(A \cap B)^c = A^c \cup B^c$.
- Axioms: $P(A) \geq 0$, $P(S) = 1$, additivity for disjoint events.
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $P(A^c) = 1 - P(A)$.

L2 · Counting

- $nPr = n!/(n - r)!$ (order matters); $nCr = n!/(r!(n - r)!)$ (selection).
- $P(A) = n(A)/n(S)$; independent choices multiply; at least = total − complement.
- Box 8R 3W 9B, 3 drawn: 14/285, 7/95, 23/57. Different suits: 169/425 = 0.3976.

L3 · Multiplication, Independence, Odds

- Independent: $P(A \cap B) = P(A)P(B)$; dependent: $P(A)P(B \mid A)$.
- Mutually exclusive + independent only if one probability is 0.
- P(at least one) = $1 - \Pi(1 - p_i)$; parallel fails if all fail.
- Odds a : b in favour $\rightarrow$ $P = a/(a + b)$.

L4 · Joint Table

- 2 × 2 table of joint probabilities; margins = totals.
- Machine A/B: cells P(machine)·P(defect | machine); P(A | D) = cell / column total.